let transformation is : X = f(Z) = aZ + b . Z ~ N(0,1) (simple base distribution).
Training Samples: $x_1 = 3$, $x_2 = -1$, $x_3 = 1$, $x_4 = 5$. Using flow based generative model, solve by hand the P(X) distribution. Write details of procedure

Thought for 72 seconds

We are given: X = f(Z) = aZ + b, with Z ~ N(0,1). We have training samples: x1=3, x2=-1, x3=1, x4=5.
We want to find the parameters a and b such that the transformed distribution fits the data.
In flow-based models, we use the change of variable formula to compute the density of X.

The change of variable formula:
p_X(x) = p_Z(z) * |det(df^{-1}/dx)|, where z = f^{-1}(x).

Since f is invertible (a linear transformation), we have:
f(Z) = aZ + b ⇒ Z = (X - b)/a.
Then, df^{-1}/dx = 1/a, so |det| = 1/|a|.

Therefore, the density of X is:
p_X(x) = p_Z((x-b)/a) * (1/|a|)

Given that Z ~ N(0,1), we have:
p_Z(z) = (1/sqrt(2π)) * exp(-z^2/2)

So:
p_X(x) = (1/sqrt(2π)) * exp(-((x-b)/a)^2/2) * (1/|a|)

We can also write this as:
p_X(x) = N(x | b, a^2) [since if Z ~ N(0,1), then aZ+b ~ N(b, a^2)]

However, note that in the flow-based model, we are using a base distribution (Z) and a transformation to model the data.
We are given 4 data points. We want to find the parameters a and b that maximize the likelihood of the data.

The log-likelihood for the data is:
L = log p_X(x1) + log p_X(x2) + log p_X(x3) + log p_X(x4)

Since the transformation is simple, we can solve for a and b by matching the moments (mean and variance) of the data.

Alternatively, we can maximize the log-likelihood directly.

Let's compute the mean and variance of the data:
data: 3, -1, 1, 5

mean = (3 -1 +1 +5)/4 = 8/4 = 2
variance = ((3-2)^2 + (-1-2)^2 + (1-2)^2 + (5-2)^2) / 4 = (1 + 9 + 1 + 9)/4 = 20/4 = 5

So the data has mean 2 and variance 5.

We know that X ~ N(b, a^2). Therefore, we set:
b = mean = 2
a^2 = variance = 5 ⇒ a = sqrt(5) or a = -sqrt(5)

Since the transformation is defined as aZ+b, and Z is standard normal, the sign of a does not affect the variance (a^2 is positive).
However, note that in the density we have 1/|a|. So both positive and negative a give the same distribution.

Let's choose a = sqrt(5). Then b=2.

Then the distribution of X is N(2, 5).

But let's verify by writing the density function:

p_X(x) = (1/sqrt(2π*5)) * exp(-(x-2)^2/(2*5))

Alternatively, using the flow formula:

p_X(x) = (1/sqrt(2π)) * exp(-((x-2)/sqrt(5))^2/2) * (1/sqrt(5))

= (1/sqrt(2π*5)) * exp(-(x-2)^2/(10))

So indeed it matches.

Therefore, the distribution of X is normal with mean 2 and variance 5.

However, note that the problem asks for P(X) distribution. We have solved it by matching moments, which is equivalent to maximum likelihood for the normal distribution.

Let's also compute the log-likelihood for the data:

log p_X(x) = -1/2 * log(2π) - 1/2 * log(5) - (x-2)^2/(10)

For the four data points:

x1=3: (3-2)^2=1 ⇒ term = -1/2*log(2π) - 1/2*log(5) - 1/10
x2=-1: (-1-2)^2=9 ⇒ term = -1/2*log(2π) - 1/2*log(5) - 9/10
x3=1: (1-2)^2=1 ⇒ term = -1/2*log(2π) - 1/2*log(5) - 1/10
x4=5: (5-2)^2=9 ⇒ term = -1/2*log(2π) - 1/2*log(5) - 9/10

Sum: 4*(-1/2*log(2π) - 1/2*log(5)) - (1/10+9/10+1/10+9/10) = 4*(-1/2*log(2π) - 1/2*log(5)) - 20/10
= 4*(-1/2*log(2π) - 1/2*log(5)) - 2

We can also compute the log-likelihood for the transformed Z variables:

z1 = (3-2)/sqrt(5) = 1/sqrt(5)
z2 = (-1-2)/sqrt(5) = -3/sqrt(5)
z3 = (1-2)/sqrt(5) = -1/sqrt(5)
z4 = (5-2)/sqrt(5) = 3/sqrt(5)

Then p_Z(z) = (1/sqrt(2π)) * exp(-z^2/2)

The log-likelihood for Z is the sum of log p_Z(z_i) and then we subtract the log|det| term for each point? Wait, careful:

In flow-based model, the log-likelihood for a data point x is:

log p_X(x) = log p_Z(z) + log |det(dz/dx)|

Here, dz/dx = 1/a, so log |det| = -log|a|.

So for each point:

log p_X(x_i) = log p_Z(z_i) - log|a|

So we can compute:

log p_Z(z1) = -1/2*log(2π) - (1/sqrt(5))^2/2 = -1/2*log(2π) - 1/10
log p_Z(z2) = -1/2*log(2π) - (9/5)/2 = -1/2*log(2π) - 9/10
log p_Z(z3) = -1/2*log(2π) - (1/5)/2 = -1/2*log(2π) - 1/10
log p_Z(z4) = -1/2*log(2π) - (9/5)/2 = -1/2*log(2π) - 9/10

Then we subtract log|a| = log(sqrt(5)) = 1/2*log(5) for each point.

So for each point:

log p_X(x_i) = [-1/2*log(2π) - (z_i^2)/2] - 1/2*log(5)

= -1/2*log(2π) - 1/2*log(5) - (z_i^2)/2

And note that (z_i^2)/2 = (x_i - b)^2/(2a^2) = (x_i-2)^2/(10)

So it matches.

Therefore, the distribution is:

X ~ N(2, 5)

And the transformation parameters are: a = sqrt(5), b = 2 (or a = -sqrt(5), but we take the positive one for simplicity).

Since the problem says "solve by hand the P(X) distribution", we can state:

p_X(x) = (1/sqrt(10π)) * exp(-(x-2)^2/10)

But note: the standard deviation is sqrt(5) so the variance is 5, and the normal density is:

p_X(x) = 1/(σ√(2π)) * exp(-(x-μ)^2/(2σ^2)) = 1/(sqrt(5)*sqrt(2π)) * exp(-(x-2)^2/(10))

So the answer is a normal distribution with mean 2 and variance 5.

> However, note that the problem says "Using flow based generative model", so we have derived it via the change of variables.
>
> We can also note that the maximum likelihood estimation for the normal distribution gives the same result.
>
> So the procedure:
>
> 1. Write the transformation and the base distribution.
> 2. Use the change of variable formula to get the density of X.
> 3. Write the log-likelihood of the data.
> 4. Maximize the log-likelihood with respect to a and b (or match moments).
>
> We did both and found:
>
> b = mean of data = 2
> a^2 = variance of data = 5
>
> Therefore, the distribution of X is N(2,5).

We are given the normalizing flow transformation:

$$X = f(Z) = aZ + b$$

with base distribution $Z \sim N(0,1)$ and 4 observed data points:

$$x_1 = 3, \quad x_2 = -1, \quad x_3 = 1, \quad x_4 = 5.$$

Step 1: Express probability density of X via change of variables

We have $Z = g(X) = \frac{X-b}{a}$, so $g'(X) = \frac{1}{a}$.

The change of variables formula for a one-to-one transformation is:

$$p_X(x) = p_Z(g(x)) \cdot |g'(x)|$$

Given $p_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$, we get:

$$p_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-b}{a}\right)^2\right] \cdot \frac{1}{|a|}.$$

So X is normally distributed:

$$X \sim N(b, a^2),$$

because $aZ + b$ with $Z \sim N(0,1)$ gives mean b and variance a^2.

Step 2: Write the log-likelihood of the data

Given data $\{x_i\}_{i=1}^4$, the log-likelihood is:

$$\mathcal{L}(a,b) = \sum_{i=1}^{4} \log p_X(x_i)$$

$$= \sum_{i=1}^{4} \left[-\frac{1}{2}\log(2\pi) - \log|a| - \frac{1}{2}\left(\frac{x_i - b}{a}\right)^2 \right].$$

Simplify (dropping constants temporarily for optimization):

$$\mathcal{L}(a,b) = -4\log|a| - \frac{1}{2}\sum_{i=1}^{4}\left(\frac{x_i - b}{a}\right)^2 + \text{const.}$$

Step 3: Maximum likelihood estimation

Let's first estimate b:

The derivative w.r.t b of $\sum(x_i - b)^2$ is:

$$\frac{\partial}{\partial b}\left[\sum(x_i - b)^2\right] = -2\sum(x_i - b) = 0.$$

So:

$$\sum_{i=1}^{4}(x_i - b) = 0 \quad \Rightarrow \quad 4b = \sum x_i.$$

$$\sum x_i = 3 + (-1) + 1 + 5 = 8 \quad \Rightarrow \quad b = 2.$$

Now estimate a^2:

Let's work with $\sigma^2 = a^2$.

We know for Gaussian MLE:

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{b})^2.$$

So:

$$(x_i - 2)^2: \quad (3-2)^2 = 1, \quad (-1-2)^2 = 9, \quad (1-2)^2 = 1, \quad (5-2)^2 = 9.$$

$$\sum (x_i - 2)^2 = 1 + 9 + 1 + 9 = 20.$$

$$\hat{\sigma}^2 = \frac{20}{4} = 5.$$

So:

$$\hat{a}^2 = 5 \quad \Rightarrow \quad |\hat{a}| = \sqrt{5}.$$

Since a is a scaling parameter and sign doesn't affect distribution of X (because Z is symmetric), we take $\hat{a} = \sqrt{5}$ (positive).

Step 4: Final distribution $p_X(x)$

$$X \sim N(\mu = 2, \sigma^2 = 5).$$

Explicitly:

$$p_X(x) = \frac{1}{\sqrt{2\pi \cdot 5}} \exp\left[-\frac{(x-2)^2}{2 \cdot 5}\right] = \frac{1}{\sqrt{10\pi}} \exp\left[-\frac{(x-2)^2}{10}\right].$$

Step 5: Verify via transformation

We can check:

$$z_i = \frac{x_i - 2}{\sqrt{5}} = \left\{\frac{1}{\sqrt{5}}, -\frac{3}{\sqrt{5}}, -\frac{1}{\sqrt{5}}, \frac{3}{\sqrt{5}}\right\}.$$

These z_i are plausible draws from $N(0,1)$, symmetric and roughly within $\pm 2\sigma$ of 0.

Final answer:

$$\boxed{N(2,5)}$$

That is:

$$p_X(x) = \frac{1}{\sqrt{10\pi}} e^{-(x-2)^2/10}.$$

Transformation is now of the form; $x_1 = a_{11}*z_1 + a_{12}*z_2 + b_1 x_2 = a_{21}*z_1 + a_{22}*z_2 + b_2$
$x_1 = 2*z_1 + 0*z_2 + 1$, $x_2 = 0*z_1 + 3*z_2 + 2$.
Let z=N(0,1). Decide on suitable training set and solve this problem using flow based model

Thought for 124 seconds

We are given a transformation from a 2D base distribution to a 2D observed variable:

$$x_1 = a_{11}z_1 + a_{12}z_2 + b_1 = 2z_1 + 0 \cdot z_2 + 1 = 2z_1 + 1,$$
$$x_2 = a_{21}z_1 + a_{22}z_2 + b_2 = 0 \cdot z_1 + 3z_2 + 2 = 3z_2 + 2.$$

Here, $\mathbf{z} = (z_1, z_2)^\top \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_2)$, i.e., z_1, z_2 independent standard normal.

The transformation is linear and invertible. In vector form:

$$\mathbf{x} = \mathbf{A}\mathbf{z} + \mathbf{b},$$

where

$$\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

The Jacobian matrix of the transformation from $\mathbf{z}$ to $\mathbf{x}$ is:

$$\mathbf{J} = \frac{\partial \mathbf{x}}{\partial \mathbf{z}} = \mathbf{A}.$$

The determinant of the Jacobian is $|\det(\mathbf{J})| = |\det(\mathbf{A})| = |2 \cdot 3 - 0 \cdot 0| = 6.$

Since the base distribution is $p_{\mathbf{z}}(\mathbf{z}) = \frac{1}{2\pi} \exp\left(-\frac{1}{2}\mathbf{z}^\top \mathbf{z}\right)$, by the change of variables formula, the probability density of $\mathbf{x}$ is:

$$p_{\mathbf{x}}(\mathbf{x}) = p_{\mathbf{z}}(\mathbf{z}) \cdot \left|\det\left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}}\right)\right| = p_{\mathbf{z}}(\mathbf{z}) \cdot \frac{1}{|\det(\mathbf{J})|},$$

where $\mathbf{z} = \mathbf{A}^{-1}(\mathbf{x} - \mathbf{b})$.

Compute $\mathbf{A}^{-1}$:

$$\mathbf{A}^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}.$$

So,

$$\mathbf{z} = \begin{pmatrix} \frac{x_1-1}{2} \\ \frac{x_2-2}{3} \end{pmatrix}.$$

Then,

$$p_{\mathbf{x}}(\mathbf{x}) = \frac{1}{(2\pi)} \exp\left(-\frac{1}{2}\left(\left(\frac{x_1-1}{2}\right)^2 + \left(\frac{x_2-2}{3}\right)^2\right)\right) \cdot \frac{1}{6}.$$

This can be written as:

$$p_{\mathbf{x}}(\mathbf{x}) = \frac{1}{12\pi} \exp\left(-\frac{1}{2}\left(\frac{(x_1-1)^2}{4} + \frac{(x_2-2)^2}{9}\right)\right).$$

This means $\mathbf{x}$ follows a bivariate normal distribution with mean $\mathbf{b} = (1, 2)^\top$ and covariance matrix $\mathbf{\Sigma} = \mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}$. Indeed, x_1 and x_2 are independent with variances 4 and 9 respectively.

Now, for training: if we want to fit a flow-based model to data generated from this distribution, we could create a training set by sampling from the base distribution and transforming, or we could assume we are given observed data drawn from $p_{\mathbf{x}}$. Let's design a suitable training set.

A "suitable training set" would be a set of samples from $p_{\mathbf{x}}$. Since we know the true parameters, we can generate samples. For example, sample $z_1, z_2 \sim \mathcal{N}(0, 1)$ and compute x_1, x_2. Let's generate a small set for illustration.

Suppose we draw 4 samples:

Sample 1: $z_1 = 0.5, z_2 = -0.2$

$$x_1 = 2(0.5) + 1 = 2, \quad x_2 = 3(-0.2) + 2 = -0.6 + 2 = 1.4.$$

Sample 2: $z_1 = -1.0, z_2 = 0.3$

$$x_1 = 2(-1) + 1 = -1, \quad x_2 = 3(0.3) + 2 = 0.9 + 2 = 2.9.$$

Sample 3: $z_1 = 0.0, z_2 = 1.1$

$$x_1 = 1, \quad x_2 = 3(1.1) + 2 = 3.3 + 2 = 5.3.$$

Sample 4: $z_1 = 1.2, z_2 = -0.8$

$$x_1 = 2(1.2) + 1 = 2.4 + 1 = 3.4, \quad x_2 = 3(-0.8) + 2 = -2.4 + 2 = -0.4.$$

So a training set of 4 points could be:

$$\mathbf{x}^{(1)} = (2, 1.4), \quad \mathbf{x}^{(2)} = (-1, 2.9), \quad \mathbf{x}^{(3)} = (1, 5.3), \quad \mathbf{x}^{(4)} = (3.4, -0.4).$$

In practice, with real data, we don't know the transformation. Here, we are pretending we only see these $\mathbf{x}$ samples and we want to learn the flow model (i.e., learn $\mathbf{A}$ and $\mathbf{b}$).

Given that the transformation is linear and the base distribution is Gaussian, the MLE for the mean and covariance of $\mathbf{x}$ is straightforward: the MLE for the mean is the sample mean, and for the covariance matrix it is the sample covariance (unbiased version divides by $n-1$, but MLE divides by n).

Let's compute from our 4 samples:

Sample mean:

$$\bar{x}_1 = \frac{2 + (-1) + 1 + 3.4}{4} = \frac{5.4}{4} = 1.35,$$

$$\bar{x}_2 = \frac{1.4 + 2.9 + 5.3 + (-0.4)}{4} = \frac{9.2}{4} = 2.3.$$

So $\hat{\mathbf{b}} = (1.35, 2.3)^\top$.

Now compute sample covariance (MLE version, dividing by 4):

First, center the data:

Centered points:
1: (0.65, -0.9)
2: (-2.35, 0.6)
3: (-0.35, 3.0)
4: (2.05, -2.7)

Covariance matrix:

$$\hat{\Sigma}_{11} = \frac{0.65^2 + (-2.35)^2 + (-0.35)^2 + 2.05^2}{4} = \frac{0.4225 + 5.5225 + 0.1225 + 4.2025}{4} =$$

$$\hat{\Sigma}_{22} = \frac{(-0.9)^2 + 0.6^2 + 3.0^2 + (-2.7)^2}{4} = \frac{0.81 + 0.36 + 9 + 7.29}{4} = \frac{17.46}{4} = 4.365.$$

$$\hat{\Sigma}_{12} = \frac{0.65 \cdot (-0.9) + (-2.35) \cdot 0.6 + (-0.35) \cdot 3.0 + 2.05 \cdot (-2.7)}{4} = \frac{-0.585 - 1.41 -}{4}$$

So MLE covariance matrix:

$$\hat{\mathbf{\Sigma}} = \begin{pmatrix} 2.5675 & -2.145 \\ -2.145 & 4.365 \end{pmatrix}.$$

But wait, the true covariance is diagonal (since $\mathbf{A}$ is diagonal). Our sample covariance is not diagonal due to small sample size. In flow-based modeling, if we assume the transformation is of the form $\mathbf{x} = \mathbf{Az} + \mathbf{b}$ with $\mathbf{A}$ being a general matrix, then we can learn $\mathbf{A}$ and $\mathbf{b}$ by maximizing the log-likelihood.

The log-likelihood for a sample $\{\mathbf{x}^{(i)}\}_{i=1}^N$ is:

$$\mathcal{L}(\mathbf{A}, \mathbf{b}) = \sum_{i=1}^N \left[\log p_{\mathbf{z}}(\mathbf{A}^{-1}(\mathbf{x}^{(i)} - \mathbf{b})) + \log|\det \mathbf{A}^{-1}|\right]$$

$$= \sum_{i=1}^N \left[-\frac{1}{2}\|\mathbf{A}^{-1}(\mathbf{x}^{(i)} - \mathbf{b})\|^2 - \frac{d}{2}\log(2\pi) + \log|\det \mathbf{A}^{-1}|\right].$$

Maximizing this is equivalent to minimizing the sum of squares of the transformed errors, but with a regularization term from the determinant.

For a linear flow with Gaussian base, the MLE of $\mathbf{b}$ is the sample mean, and the MLE of $\mathbf{\Sigma} = \mathbf{AA}^\top$ is the sample covariance matrix. However, $\mathbf{A}$ is not uniquely identified: any orthogonal transformation of $\mathbf{A}$ gives the same covariance. But if we restrict $\mathbf{A}$ to be lower triangular (for identifiability, as in normalizing flows often done), then we can use the Cholesky decomposition: $\mathbf{\Sigma} = \mathbf{LL}^\top$ with $\mathbf{L}$ lower triangular, and then take $\mathbf{A} = \mathbf{L}$. But here the true $\mathbf{A}$ is diagonal, which is a special case of lower triangular.

For our small sample, if we assume $\mathbf{A}$ is diagonal (i.e., the flow has two independent 1D flows for each dimension), then we can estimate a_{11} and a_{22} separately from the variances of each dimension. But note: if we assume diagonal $\mathbf{A}$, then we are assuming x_1 and x_2 are independent. In that case, the MLE for a_{11}^2 is the sample variance of x_1, and for a_{22}^2 the sample variance of x_2. From our sample:

Sample variance of x_1 (MLE): $\hat{\sigma}_1^2 = 2.5675$, so $\hat{a}_{11} = \sqrt{2.5675} \approx 1.602$.

Sample variance of x_2 (MLE): $\hat{\sigma}_2^2 = 4.365$, so $\hat{a}_{22} = \sqrt{4.365} \approx 2.089$.

But the true values were $a_{11} = 2$, $a_{22} = 3$. Our estimates are off due to small sample size.

If we allow a full $\mathbf{A}$, we would get $\hat{\mathbf{\Sigma}}$ as above, and then take $\hat{\mathbf{A}}$ as a matrix square root (e.g., Cholesky). The Cholesky decomposition of $\hat{\mathbf{\Sigma}}$ (if positive definite) yields a lower triangular $\mathbf{L}$ such that $\mathbf{LL}^\top = \hat{\mathbf{\Sigma}}$. Compute:

Let $\mathbf{L} = \begin{pmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{pmatrix}$. Then:

$$l_{11}^2 = 2.5675 \Rightarrow l_{11} = \sqrt{2.5675} \approx 1.602,$$

$$l_{11}l_{21} = -2.145 \Rightarrow l_{21} = -2.145/1.602 \approx -1.338,$$

$$l_{21}^2 + l_{22}^2 = 4.365 \Rightarrow l_{22}^2 = 4.365 - (-1.338)^2 = 4.365 - 1.790 \approx 2.575 \Rightarrow l_{22} \approx 1.605.$$

So $\hat{\mathbf{A}} = \mathbf{L} \approx \begin{pmatrix} 1.602 & 0 \\ -1.338 & 1.605 \end{pmatrix}$.

Then the estimated transformation is $\mathbf{x} = \hat{\mathbf{A}}\mathbf{z} + \hat{\mathbf{b}}$, with $\hat{\mathbf{b}} = (1.35, 2.3)^\top$.

This would be the flow-based model learned from the 4 samples.

Finally, the learned distribution of $\mathbf{x}$ is $\mathcal{N}(\hat{\mathbf{b}}, \hat{\mathbf{\Sigma}})$.

Thus, the procedure is:

1. Collect training data (samples of $\mathbf{x}$).
2. Assume a linear normalizing flow: $\mathbf{x} = \mathbf{A}\mathbf{z} + \mathbf{b}$, with $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$.
3. Compute MLE for $\mathbf{b}$ and $\mathbf{A}$:
   - $\hat{\mathbf{b}}$ is the sample mean.
   - $\hat{\mathbf{\Sigma}} = \frac{1}{N} \sum_{i=1}^{N} (\mathbf{x}^{(i)} - \hat{\mathbf{b}})(\mathbf{x}^{(i)} - \hat{\mathbf{b}})^\top$.
   - Take $\hat{\mathbf{A}}$ as a matrix square root of $\hat{\mathbf{\Sigma}}$, e.g., Cholesky decomposition.
4. Then the learned distribution is $p_{\mathbf{x}}(\mathbf{x}) = \mathcal{N}(\mathbf{x} | \hat{\mathbf{b}}, \hat{\mathbf{\Sigma}})$.

We demonstrated with 4 synthetic samples.

Given the initial problem statement, it only said "Decide on suitable training set and solve this problem using flow based model". So we can present the above.

Thus, the final answer can be summarized as:

We generated a training set of 4 samples from the given transformation. Then we estimated the parameters via maximum likelihood, obtaining:

$$\hat{\mathbf{b}} = \begin{pmatrix} 1.35 \\ 2.3 \end{pmatrix}, \quad \hat{\mathbf{A}} = \begin{pmatrix} 1.602 & 0 \\ -1.338 & 1.605 \end{pmatrix},$$

so the learned distribution for $\mathbf{x}$ is $\mathcal{N}(\hat{\mathbf{b}}, \hat{\mathbf{A}}\hat{\mathbf{A}}^\top)$.

Problem Setup

We have a 2D normalizing flow transformation:

$$x_1 = a_{11}z_1 + a_{12}z_2 + b_1 = 2z_1 + 1,$$
$$x_2 = a_{21}z_1 + a_{22}z_2 + b_2 = 3z_2 + 2,$$

with base distribution $\mathbf{z} = (z_1, z_2)^\top \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_2)$.

The transformation is linear:

$$\mathbf{x} = \mathbf{A}\mathbf{z} + \mathbf{b}, \quad \mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Step 1: True distribution of $\mathbf{x}$

Since $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$,

$$\mathbf{x} \sim \mathcal{N}(\mathbf{b}, \mathbf{A}\mathbf{A}^\top) = \mathcal{N}\left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}\right).$$

The exact PDF via change of variables:

$$p_{\mathbf{x}}(\mathbf{x}) = p_{\mathbf{z}}(\mathbf{z}) \cdot |\det \mathbf{J}|^{-1}, \quad \mathbf{z} = \mathbf{A}^{-1}(\mathbf{x} - \mathbf{b}),$$

with Jacobian determinant $|\det \mathbf{J}| = |\det \mathbf{A}| = 6$. Thus,

$$p_{\mathbf{x}}(\mathbf{x}) = \frac{1}{12\pi} \exp\left[-\frac{1}{2}\left(\frac{(x_1-1)^2}{4} + \frac{(x_2-2)^2}{9}\right)\right].$$

Step 2: Generate a suitable training set

We sample $\mathbf{z}$ from $\mathcal{N}(\mathbf{0}, \mathbf{I})$ and transform:

Sample	z_1	z_2	$x_1 = 2z_1 + 1$	$x_2 = 3z_2 + 2$
1	0.5	-0.2	2.0	1.4
2	-1.0	0.3	-1.0	2.9
3	0.0	1.1	1.0	5.3
4	1.2	-0.8	3.4	-0.4

Training data:

$$\mathbf{x}^{(1)} = (2.0, 1.4), \quad \mathbf{x}^{(2)} = (-1.0, 2.9), \quad \mathbf{x}^{(3)} = (1.0, 5.3), \quad \mathbf{x}^{(4)} = (3.4, -0.4).$$

Step 3: Fit a linear flow model via maximum likelihood

Assume the model: $\mathbf{x} = \mathbf{A}\mathbf{z} + \mathbf{b}$, $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The log-likelihood for N samples is:

$$\mathcal{L}(\mathbf{A},\mathbf{b}) = \sum_{i=1}^{N} \left[-\frac{1}{2}\|\mathbf{A}^{-1}(\mathbf{x}^{(i)} - \mathbf{b})\|^2 - \frac{d}{2}\log(2\pi) + \log|\det \mathbf{A}^{-1}| \right].$$

Maximizing w.r.t. $\mathbf{b}$ gives $\hat{\mathbf{b}} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}^{(i)}$.
Maximizing w.r.t. $\mathbf{A}$ leads to $\widehat{\mathbf{A}\mathbf{A}^\top} = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{x}^{(i)} - \hat{\mathbf{b}})(\mathbf{x}^{(i)} - \hat{\mathbf{b}})^\top$.

Step 3.1: Compute sample mean

$$\hat{\mathbf{b}} = \begin{pmatrix} \frac{2.0+(-1.0)+1.0+3.4}{4} \\ \frac{1.4+2.9+5.3+(-0.4)}{4} \end{pmatrix} = \begin{pmatrix} 1.35 \\ 2.3 \end{pmatrix}.$$

Step 3.2: Compute sample covariance (MLE version)

Center the data:

$$\begin{aligned} \tilde{\mathbf{x}}^{(1)} &= (0.65, -0.9), \\ \tilde{\mathbf{x}}^{(2)} &= (-2.35, 0.6), \\ \tilde{\mathbf{x}}^{(3)} &= (-0.35, 3.0), \\ \tilde{\mathbf{x}}^{(4)} &= (2.05, -2.7). \end{aligned}$$

Covariance matrix:

$$\hat{\mathbf{\Sigma}} = \frac{1}{4}\sum_{i=1}^{4}\tilde{\mathbf{x}}^{(i)}(\tilde{\mathbf{x}}^{(i)})^\top.$$

Compute elements:

$$\hat{\Sigma}_{11} = \frac{0.65^2 + (-2.35)^2 + (-0.35)^2 + 2.05^2}{4} = \frac{10.27}{4} = 2.5675,$$

$$\hat{\Sigma}_{22} = \frac{(-0.9)^2 + 0.6^2 + 3.0^2 + (-2.7)^2}{4} = \frac{17.46}{4} = 4.365,$$

$$\hat{\Sigma}_{12} = \frac{0.65(-0.9) + (-2.35)(0.6) + (-0.35)(3.0) + 2.05(-2.7)}{4} = \frac{-8.58}{4} = -$$

So:

$$\hat{\mathbf{\Sigma}} = \begin{pmatrix} 2.5675 & -2.145 \\ -2.145 & 4.365 \end{pmatrix}.$$

Step 3.3: Recover $\hat{\mathbf{A}}$ via Cholesky decomposition

Assume $\hat{\mathbf{A}}$ is lower triangular (common identifiability choice in flows). Solve $\hat{\mathbf{A}}\hat{\mathbf{A}}^\top = \hat{\mathbf{\Sigma}}$:

Let $\hat{\mathbf{A}} = \begin{pmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{pmatrix}$. Then:

$$l_{11}^2 = 2.5675 \Rightarrow l_{11} = \sqrt{2.5675} \approx 1.602,$$

$$l_{11}l_{21} = -2.145 \Rightarrow l_{21} = -2.145/1.602 \approx -1.338,$$

$$l_{21}^2 + l_{22}^2 = 4.365 \Rightarrow l_{22}^2 = 4.365 - 1.790 \approx 2.575 \Rightarrow l_{22} \approx 1.605.$$

Thus:

$$\hat{\mathbf{A}} \approx \begin{pmatrix} 1.602 & 0 \\ -1.338 & 1.605 \end{pmatrix}.$$

Step 4: Learned distribution

The estimated flow transformation is:

$$\mathbf{x} = \hat{\mathbf{A}}\mathbf{z} + \hat{\mathbf{b}},$$

with $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The implied distribution for $\mathbf{x}$ is:

$$\mathbf{x} \sim \mathcal{N}\left(\hat{\mathbf{b}}, \hat{\mathbf{\Sigma}}\right) = \mathcal{N}\left(\begin{pmatrix} 1.35 \\ 2.3 \end{pmatrix}, \begin{pmatrix} 2.5675 & -2.145 \\ -2.145 & 4.365 \end{pmatrix}\right).$$

The PDF is:

$$p_{\mathbf{x}}(\mathbf{x}) = \frac{1}{2\pi\sqrt{\det \hat{\mathbf{\Sigma}}}} \exp\left[-\frac{1}{2}(\mathbf{x} - \hat{\mathbf{b}})^\top \hat{\mathbf{\Sigma}}^{-1}(\mathbf{x} - \hat{\mathbf{b}})\right].$$

Conclusion

Using a small training set generated from the true transformation, we fitted a linear normalizing flow via maximum likelihood. Due to limited data, the estimated parameters differ from the true ones ($\mathbf{A} = \mathrm{diag}(2, 3)$, $\mathbf{b} = (1, 2)^\top$), but the procedure demonstrates the flow-based modeling approach: estimate $\mathbf{b}$ as sample mean, estimate covariance from data, and decompose it to obtain $\hat{\mathbf{A}}$.

$$\boxed{\mathcal{N}\left(\begin{pmatrix} 1.35 \\ 2.3 \end{pmatrix}, \begin{pmatrix} 2.5675 & -2.145 \\ -2.145 & 4.365 \end{pmatrix}\right)}$$